

QUBO Formulation: Alkane Structural Isomer Search

Alkane Structural Isomer Search Toy Problem

Problem Statement. Given a set of N carbon atoms and a candidate set of M possible carbon-carbon bonds, the task is to select a subset of bonds that forms a valid alkane carbon skeleton. Validity requires that each carbon atom attains a specified valence (degree), the total number of selected bonds equals $N - 1$ (ensuring a tree structure), and the selected bonds connect all carbon atoms into a single component. The objective is to find any feasible bond configuration, which corresponds to minimizing a constant objective function subject to these structural constraints.

Decision Variables. Let $\mathcal{B} = \{1, \dots, M\}$ denote the set of candidate bonds. For each candidate bond $j \in \mathcal{B}$, introduce a binary decision variable $x_j \in \{0, 1\}$, where $x_j = 1$ indicates that bond j is selected in the skeleton, and $x_j = 0$ otherwise. Each bond j connects exactly two carbon atoms, denoted $u_j, v_j \in \{1, \dots, N\}$. Let $I_i \subseteq \mathcal{B}$ be the set of candidate bonds incident to atom i , and let d_i denote the required degree for atom i .

QUBO Derivation. Step 1 — Objective. The problem is a pure feasibility problem with a constant objective function. In QUBO minimization form, this is expressed as:

$$H_{\text{obj}}(\mathbf{x}) = 0. \quad (1)$$

Step 2 — Constraints. The structural requirements are written as equality constraints:

$$\begin{aligned} \sum_{j \in I_i} x_j &= d_i, \quad \forall i \in \{1, \dots, N\}, & (\text{Degree}) \\ \sum_{j \in \mathcal{B}} x_j &= N - 1. & (\text{Bond Count}) \end{aligned}$$

The connectivity constraint requires that the subgraph induced by $\{j \in \mathcal{B} : x_j = 1\}$ is connected. For tree structures on N vertices, the degree and bond count constraints are often sufficient to enforce connectivity in small instances; we therefore focus the penalty formulation on the degree and bond count constraints.

Step 3 — Penalty Expansion. We convert each equality constraint to a quadratic penalty with weights $\lambda_D > 0$ and $\lambda_B > 0$.

(a) Squared penalty form:

$$H_D(\mathbf{x}) = \lambda_D \sum_{i=1}^N \left(\sum_{j \in I_i} x_j - d_i \right)^2, \quad (2)$$

$$H_B(\mathbf{x}) = \lambda_B \left(\sum_{j \in \mathcal{B}} x_j - (N-1) \right)^2. \quad (3)$$

(b) Full expansion using $x_j^2 = x_j$:

$$\begin{aligned} H_D(\mathbf{x}) &= \lambda_D \sum_{i=1}^N \left(\sum_{j \in I_i} x_j^2 + \sum_{\substack{j,k \in I_i \\ j \neq k}} x_j x_k - 2d_i \sum_{j \in I_i} x_j + d_i^2 \right) \\ &= \lambda_D \sum_{i=1}^N \left(\sum_{j \in I_i} (1 - 2d_i) x_j + \sum_{\substack{j,k \in I_i \\ j \neq k}} x_j x_k + d_i^2 \right), \end{aligned} \quad (4)$$

$$\begin{aligned} H_B(\mathbf{x}) &= \lambda_B \left(\sum_{j \in \mathcal{B}} x_j^2 + \sum_{\substack{j,k \in \mathcal{B} \\ j \neq k}} x_j x_k - 2(N-1) \sum_{j \in \mathcal{B}} x_j + (N-1)^2 \right) \\ &= \lambda_B \left(\sum_{j \in \mathcal{B}} (1 - 2(N-1)) x_j + \sum_{\substack{j,k \in \mathcal{B} \\ j \neq k}} x_j x_k + (N-1)^2 \right). \end{aligned} \quad (5)$$

(c) General penalty contribution to H : Since each bond j is incident to exactly two atoms u_j and v_j , the quadratic term $\sum_{\substack{j,k \in I_i \\ j \neq k}} x_j x_k$ appears in the sum for atom i only when both j and k share atom i . Thus, for any pair $j \neq k$, the coefficient of $x_j x_k$ in H_D is $2\lambda_D$ if j and k share an atom, and 0 otherwise. The linear coefficient for x_j in H_D is $\lambda_D ((1 - 2d_{u_j}) + (1 - 2d_{v_j})) = \lambda_D (2 - 2d_{u_j} - 2d_{v_j})$. For H_B , the quadratic term $\sum_{\substack{j,k \in \mathcal{B} \\ j \neq k}} x_j x_k$ expands to $2 \sum_{j < k} x_j x_k$, contributing $2\lambda_B$ to every off-diagonal entry. The linear coefficient for x_j in H_B is $\lambda_B (1 - 2(N-1)) = \lambda_B (3 - 2N)$.

Step 4 — Combined Hamiltonian.

$$\begin{aligned} H(\mathbf{x}) &= H_{\text{obj}}(\mathbf{x}) + H_D(\mathbf{x}) + H_B(\mathbf{x}) \\ &= \sum_{j \in \mathcal{B}} [\lambda_D (2 - 2d_{u_j} - 2d_{v_j}) + \lambda_B (3 - 2N)] x_j \\ &\quad + \sum_{j < k} [2\lambda_D \cdot \mathbb{I}(\text{atoms}(j) \cap \text{atoms}(k) \neq \emptyset) + 2\lambda_B] x_j x_k + c, \end{aligned} \quad (6)$$

where $c = \lambda_D \sum_{i=1}^N d_i^2 + \lambda_B(N-1)^2$.

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from the expanded Hamiltonian yields the general closed-form expressions:

$$Q_{j,j} = \lambda_D(2 - 2d_{u_j} - 2d_{v_j}) + \lambda_B(3 - 2N), \quad \forall j \in \mathcal{B}, \quad (7)$$

$$Q_{j,k} = 2\lambda_D \cdot \mathbb{I}(\text{atoms}(j) \cap \text{atoms}(k) \neq \emptyset) + 2\lambda_B, \quad \forall j \neq k, \quad (8)$$

$$c = \lambda_D \sum_{i=1}^N d_i^2 + \lambda_B(N-1)^2. \quad (9)$$

Penalty Weight Justification. The penalty weights λ_D and λ_B must be chosen sufficiently large to ensure that any constraint violation increases the energy above the minimum feasible value. Since the objective function is constant (0), the energy of any feasible solution equals the offset c . Violating any equality constraint by the minimum integer amount of 1 produces a squared penalty of at least 1. Therefore, selecting $\lambda_D, \lambda_B > 0$ guarantees that any infeasible configuration incurs a strictly positive penalty, yielding $H(\mathbf{x}) > c$. In practice, a uniform weight $\lambda = 100$ is chosen to provide a robust margin against numerical noise and to dominate any potential objective fluctuations in extended formulations.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0, 1\}^M$ recovers the optimal solution because the penalty terms are non-negative and vanish exactly when all constraints are satisfied. Consequently, the energy of any feasible solution equals the constant offset c , while any infeasible solution incurs a strictly positive penalty, yielding $H(\mathbf{x}) > c$. Therefore, the global minimum of H must occur at a feasible configuration. Since the objective function is constant, all feasible configurations are optimal, and the QUBO minimization correctly identifies the set of valid alkane skeletons.

Illustrative Example. Consider an instance with $N = 3$ carbon atoms and candidate bonds $\mathcal{B} = \{0, 1, 2\}$ corresponding to pairs (1, 2), (2, 3), and (1, 3). The required degrees are $d_1 = 1$, $d_2 = 2$, and $d_3 = 1$. Thus, $I_1 = \{0, 2\}$, $I_2 = \{0, 1\}$, and $I_3 = \{1, 2\}$. Setting $\lambda_D = \lambda_B = 100$ and noting that connectivity is redundant for $N = 3$ under these constraints, we instantiate the general formulas:

Degree penalty terms:

$$H_D(\mathbf{x}) = -400x_0 - 400x_1 - 200x_2 + 200x_0x_1 + 200x_0x_2 + 200x_1x_2 + C_D. \quad (10)$$

Bond count penalty terms:

$$H_B(\mathbf{x}) = -200x_0 - 200x_1 - 200x_2 + 200x_0x_1 + 200x_0x_2 + 200x_1x_2 + C_B. \quad (11)$$

Combining objective, penalties, and offset $c = 1000$ yields the concrete QUBO Hamiltonian:

$$H(\mathbf{x}) = -600x_0 - 600x_1 - 400x_2 + 400x_0x_1 + 400x_0x_2 + 400x_1x_2 + 1000. \quad (12)$$

Feasible configurations, such as $x = (1, 1, 0)$ (representing the linear propane skeleton), satisfy all constraints and yield $H(\mathbf{x}) = 600$. Any configuration violating degree or bond count constraints yields strictly higher energy, confirming that the global minimum correctly isolates the valid alkane isomers.